Model 1: Disruption forecast

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What it predicts and how it's scored

Model 1 fills the `disruption_prob` column: for each material and week, the probability (0 to 1) that at least one of that material's 3 suppliers is hit by an active shock that week. It is worth 18% of the total score directly, and it also tells the ordering model how much to hedge.

- **What it's checked against:** the `disrupted` column in outcomes.csv. Every week of a shock counts as 1, not only the first.
- **One supplier is enough:** lithium counts as disrupted if only one of its suppliers is hit.
- **When we forecast:** for the private leaderboard we forecast weeks 222 to 234 using data up to week 221, so 1 to 13 weeks ahead.

Scoring (Brier skill). The Brier score is the average of (prediction − actual)² over all rows; lower is better. Our score is compared with a baseline that always predicts each material's historical disruption rate:

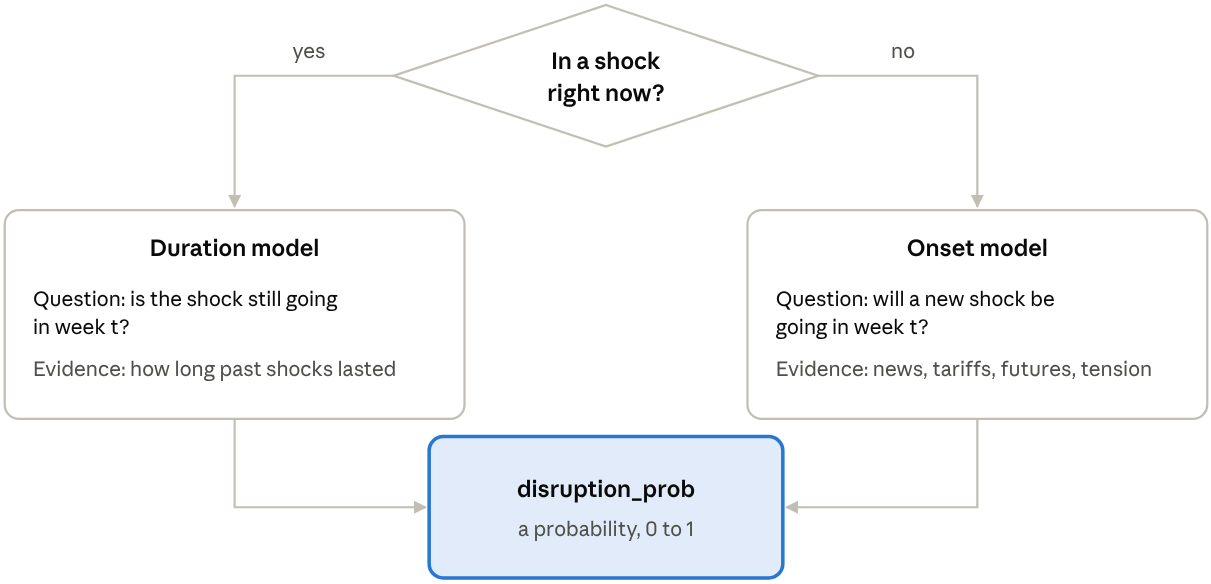
```
\text{Brier skill} = 1 - \frac{\text{our Brier score}}{\text{baseline Brier score}}
```

Matching the baseline scores 0 and a perfect forecast scores 1. Confident mistakes are expensive: predicting 0.1 when a shock happens costs 0.81 on that row, so honest, well-calibrated probabilities beat bold guesses.

The big idea: two situations, two models

At the forecast point, each material is either in a shock or calm, and those two cases need different evidence. So Model 1 is really two small models, and only one runs for each material.

Each material goes to one of two models



routing · two situations, two models

Both models output the same kind of number: the probability the material is disrupted in a future week. Neither measures how bad a shock is; severity comes from shock profiles in the ordering model.

Why not one model? "Disrupted last week" predicts "disrupted this week" so well that a single model would rely on it and learn almost nothing from the warning signs. Splitting forces the onset model to learn from warnings, which is what we need to spot new shocks, including the big one expected in the test period.

Duration model (material already in a shock)

The duration model answers: "this shock has lasted *a* weeks; what's the chance it's still going *h* weeks from now?" It uses no machine learning, just counting from past shock durations (statisticians call this survival analysis).

```
P(\text{still going at } T+h) = \frac{\#\text{ past shocks lasting longer than } a+h}{\#\text{ past shocks lasting longer than } a}
```

Worked example. Past shocks lasted 3, 4, 6, 6, 8 and 12 weeks, and lithium's current shock is in week 4. Shocks that reached 4 weeks: 4, 6, 6, 8, 12 (five). Of those, four lasted past week 6. So the chance lithium is still disrupted 2 weeks from now is about 4 ÷ 5 = 0.8.

The probability starts high and fades as you look further ahead, because shocks end:

Weeks ahead	Example probability
1	0.95
4	0.70
8	0.35
13	0.15

Two refinements:

- **Smooth the curve.** With only a few dozen past shocks, the raw counts are jumpy. Average them with a simple fitted curve (geometric or Weibull).
- **Give it a longer tail than history shows,** because the big test shock may outlast anything in training.

Onset model (material calm right now)

The onset model answers: "nothing is happening yet; how likely is a new shock to be going *h* weeks from now?" It is a machine-learning classifier, an ensemble of logistic regression and gradient boosting, that outputs a probability from 0 to 1, never a hard yes or no.

- Quiet signals (no warnings, flat futures, low tension) push the output toward the base rate, around 0.03.
- Building signals (a few reliable warnings, tension rising) push it to around 0.25.
- An imminent shock (a tariff taking effect next week, many sources confirming) pushes it to 0.7 or higher.

Training data. One row per material, origin week T and horizon h (1 to 13), only where the material was calm at T. The label is `disrupted` at week T + h. That's about 20,000 rows, but they overlap heavily, so treat the real sample as much smaller.

Features (computed at week T; country and lane signals are mapped to each material through suppliers.csv, keeping the max and mean across its 3 suppliers):

Group	Features	Source file
History	Weeks since the last shock ended; shocks in the last 52 weeks; base rate so far	outcomes
News	Warning scores over 1, 4 and 8 weeks; number of distinct sources; certainty; escalation trend	news (see next section)
Tariffs	Largest pending increase; whether its effective week falls on or before T + h	tariffs
Markets	Futures premium (futures_4w ÷ spot − 1), its 4-week change; 4-week price return	prices
Shipping	Lane congestion and its change; transit days vs normal; container rate	shipping
Macro	Tension level, 4-week change and z-score; PMI drop; fx move	macro
Supply	Lowest capacity ÷ normal; lead time above base	supply_history
Horizon	h	

Leave the material name out as a feature: with so few shocks, the model would memorize which materials got hit instead of learning warning signs.

News pipeline

News is likely the strongest early-warning signal, but some stories are rumors that never happen, so the pipeline's main job is telling real warnings from noise. It turns about 21 articles a week into roughly 10 numbers per material that feed the onset model.

1. **Tag each article.** Dictionary matching finds which materials, countries (Kestria, Valdmar, Sunda Coast, Orinoque, Halvard, Aurelia) and lanes (Merrow Strait, Northern Sea, and so on) it mentions, and which shock type (tariff, ban, sanction, strike, closure, drought, shortage).
2. **Build article features.**
    - **TF-IDF keyword vector:** scores each word or two-word phrase by how often it appears in this article versus how rare it is overall, so "export ban" scores high and "market" scores low. About 500 to 1,000 terms.

- **Sentiment:** VADER compound score. Kept only if it helps in testing, since a negative tone doesn't separate rumors from real warnings.
- **Certainty:** counts of confirming words ("announced", "effective") vs hedging words ("reportedly", "rumored", "denies").
- **Source reliability:** each source's track record (below).

3. **Train the article model.** A logistic regression predicts P(real warning) for each article. Label: an article is a real warning if an entity it mentions gets disrupted within the next 1 to 6 weeks.
4. **Add up per week and material.** Sum and max of warning scores over 1, 4 and 8 weeks, mean sentiment, and the number of distinct sources. A country mention counts toward every material that country supplies.

Source reliability score:

```
\text{reliability} = \frac{\text{warnings that came true} + 1}{\text{warnings issued} + 2}
```

The +1 and +2 keep a source with only 2 articles from scoring 100%. Also track each source's average lead time: some may warn 6 weeks ahead, others 1.

Avoiding leakage. An article's label is known only 6 weeks after it's published. When forecasting from week T, the article model and source scores may only use articles from before week T − 6. Articles *reporting* a shock that already started don't count as warnings.

Combining and calibrating

Calibration makes the probabilities honest: when the model says 0.3, about 30% of those cases should turn out disrupted. It happens in five steps.

1. **Collect out-of-fold predictions.** From the rolling backtest (next section), gather predictions the models made on data they didn't train on, with the true labels.
2. **Blend the two classifiers.** p_blend = w × p_logistic + (1 − w) × p_boosting. Try w = 0, 0.1, …, 1 and keep the lowest Brier score.
3. **Calibrate with Platt scaling,** a tiny 3-parameter logistic regression on the blended prediction (formula below). Isotonic regression is more flexible but overfits with this few shocks.
4. **Combine with the duration model.** If the material is calm: p = calibrated onset probability. If it's in a shock: p = P_cont + (1 − P_cont) × p_new, so news about a second shock still counts.

5. **Clip to [0.02, 0.98].** The floor covers no-warning policy reversals; the ceiling limits the damage from a rare false alarm.

```
\operatorname{logit}(p_{\text{cal}}) = a + b \cdot \operatorname{logit}
(p_{\text{blend}}) + c \cdot h
```

Here *a* fixes overall bias, *b* fixes confidence (above 1 means the model was too timid), and *c* lets far-ahead forecasts shrink more.

Check it with a reliability diagram: bucket predictions (0 to 0.1, 0.1 to 0.2, …) and compare the average prediction with the actual disruption rate in each bucket. Well calibrated means close to the diagonal.

Testing

Every version is tested the same way: pretend it's an earlier week, forecast the next 13 weeks using only data available then, and score against what actually happened.

- **Rolling origins:** forecast from weeks 156, 169, 182 and 195 (add 208 after Phase 2), each looking 1 to 13 weeks ahead.
- **Train only on what was known:** a training row counts only if its label week (T′ + h) is on or before the forecast origin.
- **Metric:** Brier skill against the base rate, overall, per material, and separately for calm vs in-shock rows.
- **Fit the blend weight and calibrator on earlier folds only** (for example, fit on 156 to 182 and test on 195).

Ablation ladder. Add one piece at a time and keep it only if Brier skill improves. This also makes a good chart for the video.

Version	Adds	Expected result
v0	Base rate	Skill = 0 by definition
v1	Duration model (persistence)	Probably the biggest single gain
v2	Tariff features	Catches announced hikes
v3a	News: tags + keyword counts	First news signal
v3b	+ TF-IDF vector	

Version	Adds	Expected result
v3c	+ sentiment	Keep only if it helps
v3d	+ source reliability + certainty	Should filter rumors
v4	Markets, shipping, macro, ensemble, calibration	Full model

Outputs for the ordering model, and the big shock

Besides the submission column, Model 1 tells the ordering model which suppliers and lanes are at risk, because a ban on one country and a strait closure call for different responses.

- **Supplier labels** from supply_history: a supplier counts as hit when capacity drops below about 70% of normal, lead time rises 1 week or more above base, or landed cost jumps beyond what the price change explains.
- **Lane labels:** every country on a lane hit in the same week (a Merrow Strait closure hits both Kestria and Sunda Coast).
- **Run the same two-part model** at country and lane level to get hazards for the scenario generator. Each supplier has its own state, so news about a second shock still counts while another supplier is already hit.
- **Combine to material level** as a check: P(material disrupted) ≈ $1 - (1 - p_1)(1 - p_2)(1 - p_3)$.
- **Quick version:** split the material probability across suppliers in proportion to their risk features.

Preparing for the big shock. We were hinted that the test period holds a shock larger than anything in training.

- Monotonic constraints on the boosting model and the logistic member keep probabilities rising as signals go past their training ranges.
- When Phase 2 data lands, compare signals in weeks 209 to 221 with the training maximums. If they exceed them, lean on the logistic member or apply a deliberate uplift after calibration, only where the signals are, and write down what you changed.
- Don't raise probabilities across the board: materials that stay calm would cost Brier points.

Build order

Build the backtest loop first, so every step is measured as soon as it exists. Each step leaves a working column behind.

- [ ] Rolling backtest loop and Brier skill scoring (30 min)
- [ ] Episode table and duration model → v1 (30 min)
- [ ] Tariff features → v2 (1 hr)
- [ ] News tagging, TF-IDF, sentiment, source scores, article model → v3 (2 to 3 hr, overlaps with the event study)
- [ ] Market, shipping and macro features, ensemble, calibration → v4 (2 hr)
- [ ] Supplier and lane hazards for the ordering model (2 hr)

Open question for the event study: do tariff hikes count as `disrupted = 1`? If yes, announcements make those weeks almost free to predict. If no, tariffs only matter for costs in the ordering model.